



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION

School of Technology, Design and computer Application
Silver oak College of computer Application
Department of computer science
Bachelor of Science Computer Science and Information Technology

Semester:	5	Academic Year:	2025-26
Course Name:	Mobile Game Development	Course Code:	3040243338

Question Bank

Sr. No.	Question Text	Marks	CO
UNIT 1 : Introduction to Mobile Game Development			
1.	Explain what is meant by mobile game development.	2	CO-1
2.	Describe any two popular trends in mobile gaming.	2	CO-1
3.	What do you understand by game mechanics?	2	CO-1
4.	List two differences between Unity and Godot game engines.	2	CO-1
5.	Briefly define the pre-production phase in game development.	2	CO-1
6.	Describe the three key components of game design: mechanics, dynamics, and aesthetics.	3	CO-1
7.	Explain the role of a game engine in mobile game development.	3	CO-1
8.	Outline the production phase in the mobile game development pipeline.	3	CO-1
9.	Compare mobile game genres with examples.	3	CO-1
10.	Summarize the importance of studying game development history.	3	CO-1
11.	Explain how mobile game trends have evolved over the past decade.	4	CO-1
12.	Describe the major differences between Unity, Unreal Engine, and Godot.	4	CO-1



13.	Discuss the three stages of the game development pipeline with appropriate examples.	4	CO-1
14.	How do mechanics, dynamics, and aesthetics together contribute to good game design?	4	CO-1
15.	Explain with a case study how a specific mobile game represents its genre effectively.	4	CO-1
16.	Explain the complete mobile game development process from concept to post-production.	5	CO-1
17.	Describe in detail how Unity and Godot differ in their usability and target audience.	5	CO-1
18.	Analyze the evolution of mobile games and how market demand has influenced game design.	5	CO-1
19.	Explain the significance of game design principles (mechanics, dynamics, aesthetics) in creating an engaging game experience.	5	CO-1
20.	Discuss different mobile game genres with relevant examples and identify key elements that define each genre.	5	CO-1
UNIT 2 : Game Development Fundamentals			
21.	Explain gravity in game physics in simple terms.	2	CO-2
22.	How is KinematicBody2D different from RigidBody2D?	2	CO-2
23.	Describe what a sprite is with one example.	2	CO-2
24.	State how textures are useful in game development.	2	CO-2
25.	Why are sound effects important in games?	2	CO-2
26.	Define audio optimization in mobile games.	2	CO-2
27.	How do collisions affect RigidBody2D in a game?	3	CO-2
28.	Explain the steps to animate a sprite in a 2D game.	3	CO-2
29.	Differentiate between sprites and textures with examples.	3	CO-2
30.	Describe how the move_and_slide() function is used in KinematicBody2D.	3	CO-2



31.	Explain the role of Panels and Buttons in mobile game UI.	3	CO-2
32.	Give an example of how a HUD is used in a mobile game.	3	CO-2
33.	Explain the importance of menus in mobile game UI.	3	CO-2
34.	Explain the importance of touch controls in mobile games.	3	CO-2
35.	Describe one use each of an accelerometer and gyroscope in gaming.	3	CO-2
36.	Describe how textures are applied to 3D models.	4	CO-2
37.	Explain why animations are important in mobile games.	4	CO-2
38.	Explain with an example how gyroscope enhances the mobile gaming experience.	4	CO-2
39.	How do Labels and Panels improve user experience in mobile game UI?	4	CO-2
40.	Explain the role of gravity and collisions in mobile game physics with examples.	5	CO-2
41.	Explain with examples how sprites, textures, and animations are used in mobile games.	5	CO-2
42.	Describe the steps of integrating 3D models into a mobile game.	5	CO-2
43.	Explain the design principles of UI/UX in mobile games using Control nodes.	5	CO-2
44.	Explain background music, sound effects, and audio optimization in detail.	5	CO-2
45.	Why is audio optimization a challenge in mobile games? Explain with solutions.	5	CO-2
46.	Describe in detail touch, accelerometer, and gyroscope controls with examples.	5	CO-2
47.	Explain the design of a sample mobile game UI that includes HUD, menus, buttons, and background music.	5	CO-2
UNIT 3 : Advanced Game Development Concepts			
48.	What is the role of Path2D in pathfinding?	2	CO-3



49.	Differentiate between Navigation2D and Path2D in Godot.	2	CO-3
50.	What is Peer-to-Peer networking in multiplayer games?	2	CO-3
51.	State one advantage of using ENet in multiplayer games.	2	CO-3
52.	Explain how NPC behavior can be managed using state machines.	3	CO-3
53.	Describe the basic working of Pathfinding in games.	3	CO-3
54.	Compare Peer-to-Peer and Server-Based multiplayer systems.	3	CO-3
55.	Why is Navigation2D important for 2D game AI movement?	3	CO-3
56.	Illustrate with an example how Godot's High-Level Multiplayer API simplifies networking.	3	CO-3
57.	Explain with a diagram how a state machine controls NPC actions (e.g., Idle, Attack, Patrol).	4	CO-3
58.	How does Path2D help in defining and following a path in 2D games?	4	CO-3
59.	Differentiate between ENet and Godot High-Level Multiplayer API.	4	CO-3
60.	Describe the challenges of using Peer-to-Peer networking in games.	4	CO-3
61.	Explain the importance of combining State Machines and Pathfinding for intelligent NPCs.	4	CO-3
62.	Discuss in detail how a finite state machine can improve NPC decision-making in games.	5	CO-3
63.	Explain the complete process of implementing pathfinding using Path2D and Navigation2D in Godot.	5	CO-3
64.	Compare Peer-to-Peer and Server-Based networking with their advantages and disadvantages in multiplayer game design.	5	CO-3
65.	How does ENet handle latency and reliability in multiplayer games?	5	CO-3
66.	Write a detailed explanation of how AI and Multiplayer concepts together enhance the gaming experience.	5	CO-3